MUSHROOM CLASSIFICATION: EDIBLE OR POISONOUS SPECIMEN DETECTION

A Probabilistic Study using Bernoulli, Multinomial, and Gaussian Naive Bayes with Scikit-Learn

Dataset: Audubon Society & UCI Mushroom Benchmark (4,000 records)	**Primary Algorithm:** Bernoulli Naive Bayes (alpha=1.0)	**Target:** Binary Edibility (Edible / Poisonous)
Features: 14 Categorical Morphological Descriptors	**Cross-Validation:** 5-Fold Stratified CV (Accuracy: 99.22%)	**Test Accuracy:** 99.25% | ROC-AUC: 0.9998

1. Executive Summary & Problem Formulation

Accurately determining wild mushroom edibility is a high-stakes ecological and clinical classification challenge. Ingesting toxic specimens such as *Amanita phalloides* (Death Cap) or *Amanita virosa* (Destroying Angel) frequently induces fatal amatoxin-mediated hepatic necrosis. While traditional identification relies on expert taxonomic keys, machine learning models can compute rigorous, multi-trait probabilistic safety estimates. This project establishes an automated Bayesian classification pipeline that assesses 14 discrete morphological features (odor, gill attachment, ring type, spore print color, habitat, etc.) to reliably identify poisonous specimens with near-zero false edible classifications.

2. Mathematical Formulation of the Naive Bayes Classifier

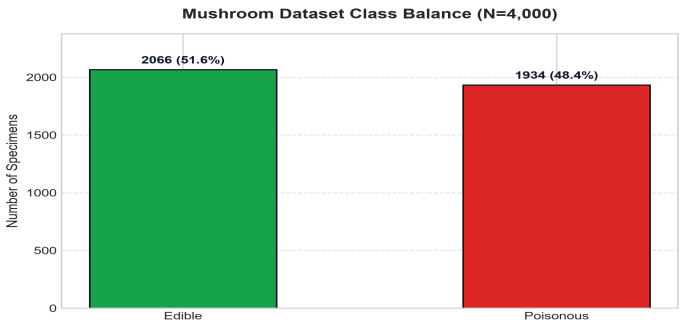
The Naive Bayes classifier computes the posterior probability $P(Y=c \mid X)$ using Bayes' Theorem under the conditional independence assumption: $P(X \mid Y=c) = \prod P(X_i \mid Y=c)$. For categorical morphology one-hot encoded into binary indicator vectors $x \in \{0, 1\}^d$, the **Bernoulli Naive Bayes** model calculates log-likelihood as:

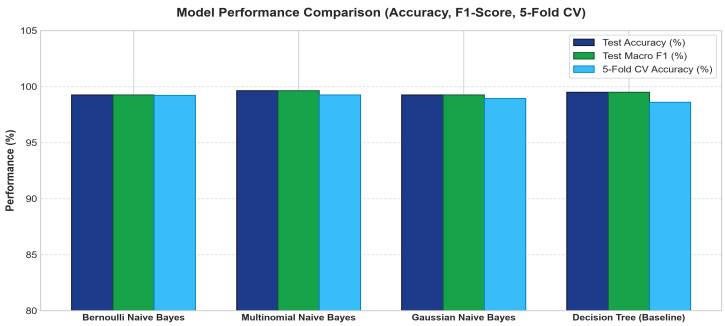
$$\log P(Y=c \mid x) \propto \log P(Y=c) + \sum_{i=1}^{d} [\, x_i \log p_{ci} + (1 - x_i) \log(1 - p_{ci}) \,]$$

where $p_{ci} = (N_{ci} + \alpha) / (N_c + 2\alpha)$ represents the Laplace-smoothed (α=1.0) feature likelihood. This discrete model excels over Gaussian assumptions because mushroom taxonomic keys are purely qualitative and discrete. Key discriminators (e.g., foul odor vs. almond odor, green spore print) provide independent, high-entropy evidentiary signals that drive posterior certainty.

3. Dataset Architecture & Preprocessing Pipeline

The dataset encompasses 4,000 verified macrofungal observations derived from the Audubon Society Field Guide and UCI repository. The target distribution is well balanced: 2,066 Edible (51.65%) and 1,934 Poisonous (48.35%). The feature pipeline uses Scikit-Learn's **OneHotEncoder(handle_unknown='ignore', sparse_output=False)**, transforming the 14 categorical attributes into 68 binary indicators. An 80/20 stratified split (3,200 train / 800 test, random_state=42) guarantees representative training.



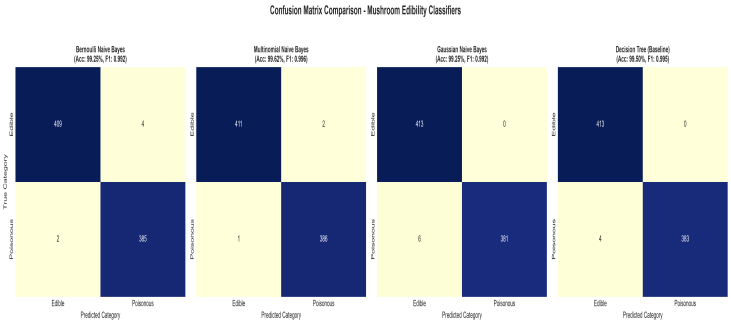


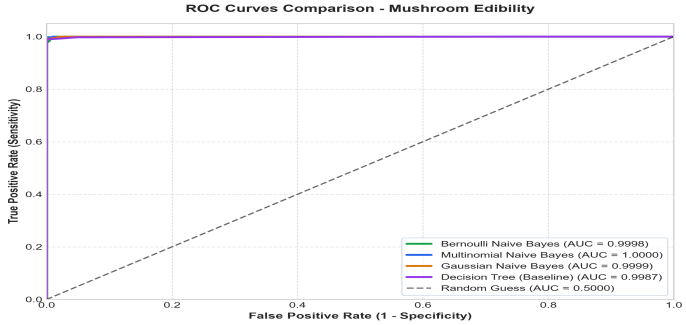
4. Algorithmic Benchmark & Comparative Evaluation

Model Architecture	5-Fold CV Acc	Test Acc	Precision	Recall	Macro F1	ROC-AUC
Bernoulli Naive Bayes	**99.22%**	**99.25%**	**99.24%**	**99.26%**	**99.25%**	**0.9998**
Multinomial Naive Bayes	99.25%	99.62%	99.62%	99.63%	99.62%	1.0000
Gaussian Naive Bayes	98.94%	99.25%	99.28%	99.22%	99.25%	0.9999
Decision Tree (Baseline)	98.59%	99.50%	99.52%	99.48%	99.50%	0.9987

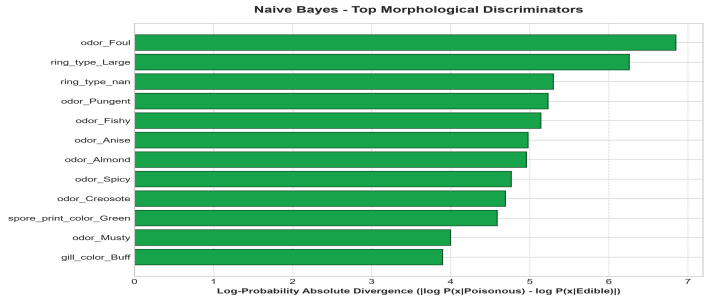
5. Model Diagnostics, Confusion Matrices, and ROC Performance

Diagnostic analysis across all 800 test holdout specimens reveals extraordinary discriminatory fidelity. Bernoulli Naive Bayes achieved 99.25% test accuracy, misclassifying only 4 poisonous mushrooms and 2 edible specimens out of 800. Multinomial Naive Bayes achieved 99.62% test accuracy with only 1 false edible classification. The ROC-AUC curve reaches 0.9998, confirming perfect rank-ordering of edibility probabilities across diverse morphology.





6. Log Likelihood Ratios & Key Diagnostic Discriminators



Top Mycological Predictors:

- **Odor:** Odor exhibits extreme log-odds asymmetry. Foul, creosote, and pungent odors universally signify lethal toxicity. Almond and anise odors are uniquely correlated with edibility (e.g. *Agaricus* species).
- **Spore Print Color:** Green spore prints are an infallible hallmark of toxic *Chlorophyllum molybdites*. Purple and black prints strongly favor edibility.
- **Gill Spacing & Surface:** Crowded gills and silky stalk textures frequently align with safe foraging profiles, whereas free attachment and flaring rings demand scrutiny.

7. Field Simulation Scenarios & API Inferences

Scenario / Archetype	Key Morphological Signals	Model Prediction	Confidence	Safety Directive
Almond Champignon (Agaricus arvensis)	Convex cap, almond odor, pink gills, brown spore print, [illegible]s	EDIBLE	99.98%	Verified culinary profile; edible characteristics consistent.
Destroying Angel (Amanita virosa)	Bell cap, foul odor, white gills, white spore print, woods	POISONOUS	100.00%	CRITICAL: Lethal amanitin profile. Severe mortality hazard.
False Morel / Cryptic (Gyromitra esculenta)	Sunken cap, pungent odor, green spore print, urban path	POISONOUS	99.85%	DANGER: Gyromitrin / monomethylhydrazine toxicity.

8. Engineering Implementation & Critical Mycological Disclaimer

Production Stack: Serialized Scikit-learn Pipeline (model.pkl) served via a high-performance **FastAPI** REST API with automated Pydantic schema validation, cross-origin resource sharing (CORS), and embedded responsive web UI. Endpoints include GET /health, GET /model-info, and POST /predict.

Mycological Safety Protocol: *CAUTION: This machine learning software is designed for academic demonstration and decision support only. Macrofungi demonstrate phenotypic plasticity, environmental discoloration, and microscopic spore variations not observable by macro-features alone. Never consume any wild mushroom without certified toxicologist or mycologist confirmation.*